

When running numbers that were relatively small, for example, 5,2; 9, 3, etc. I noticed that the model did not have any difficulty generating results. When generating bigger numbers, I noticed that the model made more errors and was getting more frustrated with each error. After counting the number of digits of each correct and incorrect answer, I deduced that the model fails when the answer contains more than 8 digits.

After calculating the answers myself, I deduced that the answers were mostly wrong in the middle digits. The model has a good sense of roughly how big the answer is and what it ends with, but it can't work out the middle. The first comes from estimating the size, and the last can follow a pattern (anything ending in 6, when squared, ends in 6). The model is producing something that looks like a correct answer rather than doing the math.

GPT-4.1 nano fails because it doesn't actually calculate; it predicts the most likely result based on token patterns from its training data. Small multiplications like 99×99 appear so often online that the model has basically memorized them, which explains why it had no trouble with smaller numbers. However, a number with 17 digits squared has most likely never been written anywhere, so the model has nothing to remember and ends up guessing. I also learned that the model does not read numbers the way we do. It splits them into chunks of up to three digits, so "221533456" becomes "221 | 533 | 456" instead of one number. Since multiplication requires carrying digits from right to left, but the model writes its answer from left to right in chunks, it loses track of the middle. This also explains my strangest results, where the model gave up completely and repeated the same three digits over and over, like "747" and "174", until the answer was much longer than it should have been.

Overall, this experiment showed me that GPT-4.1 nano is great with words but cannot be trusted with math without a calculator.